

MicroPulse — Composite Load Guide

Reading the single-number training load summary

MicroPulse Performance Intelligence Platform

April 2026

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1. What this is

Composite Load is MicroPulse’s attempt to answer a single coach question with a single number: **“how hard did this player work overall?”**

A player’s training session generates a dozen separate signals — RPE, total distance, sprint count, deceleration count, metabolic power, accel/decel ratio, heart-rate exposure, and so on. Each one tells you something useful in isolation. None of them tell you the whole story. A player can have low GPS distance (technical session) but high RPE (because the technical session was demanding mentally), or high GPS distance with low RPE (because they jogged through a recovery session). Reading any single signal misleads you in those cases.

The Composite Load score blends multiple signals into a single 0-1 number per player per day, with **explicit weights** based on which signals are most predictive of injury risk and adaptation in the published sport-science literature. You can scan 18 composite scores in 30 seconds and find the players who deserve a deeper look — instead of opening 18 individual signal dashboards.

There are TWO composite scores in MicroPulse, not one. - Composite Load Concern (the one you see as “Comp 0.78” in Daily Briefing) — for outdoor/mixed sessions. - **Composite Indoor Load Score** (the one on the Indoor Load page, scaled 0-200) — purely IMU-driven, for hall sessions where GPS is unreliable.

They use different formulas because the underlying signals differ. The same player can show “Comp 0.40” outdoor and “Composite 142” indoor on the same day — both can be correct, they answer slightly different questions.

2. Composite Load Concern (outdoor / mixed) — the “Comp 0.78” number

This is the one you see in the Daily Briefing top concerns list and in the Decision Summary card. Score range is 0.00 (no concern) to 1.00 (very high concern).

2.1 What goes into it

Up to seven signals contribute, each on its own 0-1 sub-scale. They are blended together with weights tuned to published injury-risk evidence:

Signal	Weight (when present)	What it measures
RPE-based ACWR	55%	Internal load (perceived effort × duration)
GPS Neuromuscular Burden Score	45%	External load (sprint, decel, HSR exposure)
Metabolic Load Score	added when available	Aerobic / metabolic demand (separate from mechanical)
Decel Burden Score	bumps when very high	High-intensity braking specifically
Residual MLI	safety-net	3-day rolling Mechanical Load Index
Residual Decel Index	safety-net	3-day rolling decel exposure
HID% Fatigue Flag	bump	When player can’t reach high speeds anymore

When only some signals are available (e.g. no Catapult sync today), the score is computed from what’s there with a confidence discount. Better to have an estimate than no signal at all.

2.2 Why those weights

The 55/45 split between internal and external is from the published consensus on training load monitoring (Halsen 2014; Bourdon et al. 2017): both matter, internal slightly more because it captures how the player is *perceiving* the work. The other signals enter as either additive context (metabolic, decel) or safety-net thresholds (residual MLI / decel — a player whose 3-day accumulated load is in CAUTION or HIGH is bumped to at least “moderate” concern, regardless of today’s session).

Why not just use the highest single signal?

Because high single signals are common and not always concerning. A striker who sprints 25 times in a session has a high sprint count — but if their RPE was moderate and their ACWR is in the sweet spot, this is just a normal striker session, not a flag. The composite avoids over-flagging by requiring multiple signals to agree before raising concern.

2.3 Reading the score

Composite Score

0.00 – 0.14

0.15 – 0.39

0.40 – 0.67

0.68 – 1.00

When you see “Comp 0.78” in a player row, the player is in the **High** band — that’s why they appear in the Top Concerns list with a red dot.

2.4 The Daily Briefing chip

In the Daily Briefing top-concerns list, each player gets a row like:

“Andri Rafn Yeoman · Score 20/25 · Comp 0.80 · PL 1.99× · mechanical fatigue”

- **Score 20/25** — wellness self-report (sleep + soreness + readiness summed)
- **Comp 0.80** — the Composite Load score we’re explaining (high)
- **PL 1.99×** — Player Load multiplier vs personal baseline (1.99× = nearly twice their normal)
- **mechanical fatigue** — categorical fatigue type derived from which signals dominated the composite

You can read this in 1 second once you know the chips. “Composite high + PL 2x baseline + mechanical fatigue” = player just had a session that was much harder than normal AND the dominant fatigue is mechanical (decel, sprints, accelerations) rather than metabolic.

2.5 Fatigue type

When both Mechanical Load Index and Metabolic Load Score are available, the engine also categorises the fatigue:

- **mechanical_fatigue** — decels and sprints dominant (most common in defenders, central midfielders)
- **metabolic_fatigue** — sustained running, HSR, high HMLD dominant (most common in wingers, attackers)
- **global_fatigue** — both dominant (typically post-match or hard friendly)
- **normal** — neither dominant

This tells you not just *that* the player is tired, but *what kind of tired* — which informs the modification. Mechanical fatigue → drop the decel-heavy drill. Metabolic fatigue → drop the long-running drill. Global fatigue → cut the whole session intensity.

3. Composite Indoor Load Score — the “Composite 142” number

This is a completely different formula used on the **Indoor Load** page. Scale runs roughly 0-200, with 100 = the player’s personal baseline (not zero like the outdoor composite).

3.1 Why a different formula

Indoor sessions break GPS in ways that make the outdoor composite unreliable. Velocity bands don’t resolve well, HMLD becomes noisy, sprint count under-reports. The indoor composite was built specifically without GPS-derived signals so it stays accurate when the team is in the hall.

3.2 What goes into it

Three IMU-only signals:

Signal

Player Load

Dynamic-High % time

IMA total

All three are accelerometer-derived (not GPS), so they work indoors. Each is normalised against the player’s personal 28-day indoor baseline before blending.

3.3 Reading the score

Score range

< 60

60 – 89

90 – 110

111 – 140

> 140

A score of 142 means *“this session was 42% above what this player typically does indoors.”*

3.4 Why 100 is normal, not 50

Because the indoor composite is normalised against each player’s own baseline, 100 means “exactly your average”. Above 100 means harder than normal, below 100 means easier than normal. This is the opposite of the outdoor composite (where 0 is normal, 1 is concerning), and that difference can confuse coaches who use both.

A handy rule of thumb: - Outdoor “Comp 0.7+” → flag - Indoor “Composite 140+” → flag

Different scales, same coach action.

4. Putting them together — a worked example

Let's say Jónatan Guðni Arnarsson trained at the indoor hall yesterday and you open his Decision Summary modal in the morning.

You see: - **Daily Briefing chip**: "Comp 0.42 · PL 1.6×" (outdoor composite says: borderline moderate) - **Indoor Load row**: "Composite Score 152 · Spike" (indoor composite says: significantly harder than his normal) - **Wellness check-in**: 17/25 (slept 3, sore 4, readiness 4 — slightly below normal) - **AI Summary**: *"Yesterday's hall session was much harder than Jónatan's typical. He's reporting slight fatigue this morning. No restrictions today, but consider dropping the decel finishing drill — most of yesterday's spike came from braking work."*

What's happening: indoor composite caught a real spike (the IMU saw it). The outdoor composite is computed from the same data with different weighting and lands at "moderate" because it includes ACWR which smooths over a chronic baseline. Both are right; the indoor one is more sensitive to single-day intensity, the outdoor one to multi-day patterns.

The coach action is the same: read the AI summary's recommendation, drop the decel drill, retest tomorrow.

5. Common patterns

5.1 High composite + low wellness self-report

Two signals agreeing → high confidence the player needs modification. This is the cleanest "act now" signal in the platform.

5.2 High composite + normal wellness

Player did a hard session but feels fine. Two reads: - **If post-match (MD+1, MD+2)** → expected. Coach Guidance will say no action, AI Summary annotates "post-match echo". - **If post-training day** → either the player is exceptionally fit (high adaptation reserve) or they're under-reporting fatigue. Watch them for 24-48h. If wellness drops tomorrow, they were under-reporting.

5.3 Normal composite + low wellness

Player feels tired but training data says normal load. Almost always non-training cause: poor sleep, life stress, illness incoming, hangover, period-related, etc. The wellness self-report is the more reliable signal here — modify based on that.

5.4 Normal composite + normal wellness

Train as planned. No further investigation needed.

5.5 Indoor composite “Spike” + outdoor composite “Low”

You’re looking at a player whose indoor session was much harder than usual, but the chronic outdoor data dominates the outdoor composite. Common during winter when the team has shifted to indoor. Trust the indoor composite — it’s more current.

6. What it does NOT capture

Honest limits of the Composite Load score:

- **Position-specific demands** — a centre-back’s high decel load is a normal day; a striker’s same load is a hard day. The personal baselines partially address this but not fully. Use position context when reading.
 - **Mental / off-field load** — if the player had a child up sick all night, the composite knows nothing about it. The wellness check-in is the only signal that captures this.
 - **Acute injuries** — composite tracks load, not pain. A pulled hamstring at minute 60 might not show in the load numbers because the session was cut short. Always pair with the injury / availability log.
 - **Chronic vs acute distinction** — the outdoor composite includes ACWR, which IS chronic-aware. The indoor composite is more session-focused.
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7. Workflow — how to use composite score in a normal week

7.1 Daily — morning briefing

Open the Daily Briefing. Sort by Top Concerns (already sorted by composite descending). For each player above 0.7:

1. Read the row — composite, PL multiplier, fatigue type
2. Open Decision Summary modal
3. Read the AI Summary — it synthesizes composite + pattern + match context
4. Decide modification

For players in the 0.4–0.7 range, just glance at the Daily Briefing — no modal needed unless something else (wellness drop, ACWR spike) also fires.

7.2 Weekly — periodisation review

Look at the team-average composite over the week. A team consistently around 0.3–0.5 is in a healthy training rhythm. Several days above 0.6 means you’re pushing the squad — make sure the next week has recovery built in.

7.3 Returning from injury

For a player coming back from injury, composite under 0.2 for the first 2 weeks is healthy — they're being protected. Watch the trajectory: if composite rises 0.1 per week, that's a controlled return. If it jumps from 0.1 to 0.5 in one session, you progressed too fast.

8. Troubleshooting

“Composite shows 0.00 for half my squad”

GPS data missing for those players today. The composite degrades gracefully — if no signals are available it returns 0.00 with a “none” concern level. Open /coach/load and check if the Catapult sync ran for those players. Most common cause: the player was out for the session (so no pod data). Less common: pod malfunction or sync failure.

“Composite says High but the player feels fine and trained well”

See Section 5.2. The most likely explanation is post-match echo — check the Daily Briefing reason chip, it should now say “(post-match — expected)” if today is MD+1 or MD+2. If it doesn't, the player is genuinely well-adapted to the load they're carrying. Trust the player's body but watch for delayed fatigue tomorrow.

“Indoor and outdoor composite disagree”

Different formulas, different signals — they're answering different questions. See Section 4 for a worked example. When in doubt, trust whichever composite is closer to the data the player actually generated today. Indoor session today → trust indoor composite. Outdoor session → trust outdoor composite.

“Composite stays at 0.0 even on hard match days”

You probably don't have RPE check-ins flowing OR your team isn't using ACWR yet. The internal-load arm of the composite needs daily session_rpe entries. Check the Compliance row in the Daily Briefing — if RPE submission is consistently at 0/N, that's why.

Last updated: 29 April 2026 · MicroPulse Performance Intelligence Platform · Methodology: Halson 2014 (training load monitoring); Bourdon et al. 2017 (load monitoring consensus); Hulin et al. 2014 (ACWR); Gabbett 2016 (training-injury prevention); McBurnie et al. 2022 (deceleration framework); Harper et al. 2019 (HID% fatigue)